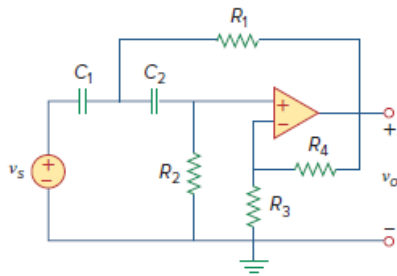


Problem

In the op amp circuit of Fig. 10.118, find the closed-loop gain and phase shift of the output voltage with respect to the input voltage if $C_1 = C_2 = 1 \text{ nF}$, $R_1 = R_2 = 100 \text{ k}\Omega$, $R_3 = 20 \text{ k}\Omega$, $R_4 = 40 \text{ k}\Omega$, and $\omega = 2000 \text{ rad/s}$.

Figure 10.118:



Step-by-step solution

Step 1 of 6

Given data,

$$C_1 = C_2 = 1 \text{ nF}$$

$$R_1 = R_2 = 100 \text{ k}\Omega$$

$$R_3 = 20 \text{ k}\Omega$$

$$R_4 = 40 \text{ k}\Omega$$

$$\text{and } \omega = 2000 \text{ rad/s}$$

Step 2 of 6

In frequency domain

$$C_1 \rightarrow \frac{1}{j\omega C_1}$$

$$C_1 \rightarrow \frac{1}{j \times 2000 \times 1 \times 10^{-9}}$$

$$C_1 \rightarrow -j5 \times 10^5 \Omega$$

And

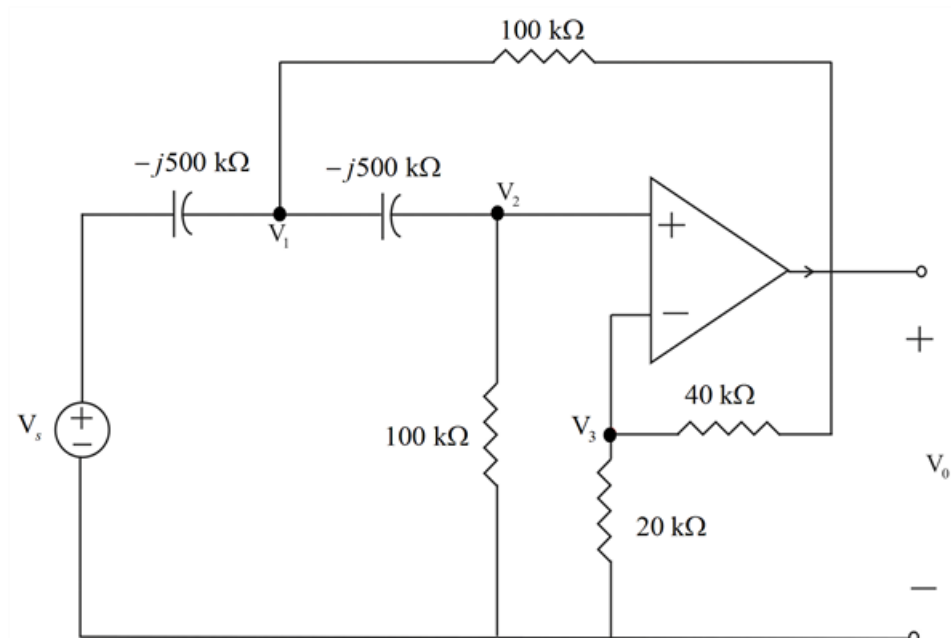
$$C_2 \rightarrow \frac{1}{j\omega C_2}$$

$$C_2 \rightarrow \frac{1}{j \times 2000 \times 1 \times 10^{-9}}$$

$$C_2 \rightarrow -j5 \times 10^5 \Omega$$

Step 3 of 6

Now the circuit is,



Step 4 of 6

By applying KCL at node V_1 , we get

$$\frac{V_1 - V_s}{-j500 \times 10^3} + \frac{V_1 - V_2}{-j500 \times 10^3} + \frac{V_1 - V_o}{100 \times 10^3} = 0$$

$$\frac{V_1 - V_s}{-j5} + \frac{V_1 - V_2}{-j5} + \frac{V_1 - V_o}{1} = 0$$

$$V_1(1 + j0.4) + V_2(-j0.2) - V_o + V_s(-j0.2) = 0$$

$$V_1(1 + j0.4) + V_2(-j0.2) - V_o = V_s(j0.2) \quad \dots\dots (1)$$

Step 5 of 6

By applying KCL at node V_2 , we get

$$\frac{V_2 - V_1}{-j500 \times 10^3} + \frac{V_2}{100 \times 10^3} = 0$$

$$V_2 \left(\frac{1}{-j500 \times 10^3} + \frac{1}{100 \times 10^3} \right) = \frac{V_1}{-j500 \times 10^3}$$

$$V_2 \left(\frac{1}{-j5} + \frac{1}{1} \right) = \frac{V_1}{-j5}$$

$$V_1 = V_2(1 - j5) \quad \dots\dots (2)$$

Step 6 of 6

From the figure,

$$V_3 = V_o \times \frac{20 \times 10^3}{20 \times 10^3 + 40 \times 10^3}$$

$$V_3 = \frac{V_o}{3}$$

For an ideal op amp,

$$V_2 = V_3$$

$$V_2 = \frac{V_o}{3}$$

By substituting the above value in (2) we get,

$$V_1 = \frac{V_o}{3}(1 - j5)$$

By substituting the above value in (1) we get,

$$\frac{V_o}{3}(1 - j5)(1 + j0.4) + \frac{V_o}{3}(-j0.2) - V_o = V_s(j0.2)$$

$$V_s(j0.2) = \frac{V_o}{3}[3 - j4.6 - j0.2 - 3]$$

$$V_s = \frac{V_o}{3} \frac{[0 - j4.8]}{j0.2}$$

$$\frac{V_o}{V_s} = -0.125$$

$$\therefore \boxed{\frac{V_o}{V_s} = 0.125 \angle 180^\circ}$$